NOTRE DAME UNIVERSITY
BANGLADESH

Machine Learning Lab Report-05

Course Code: CSE4214

Course Title: Machine Learning Lab

Lab Task Topic: Naive Bayes Theorem

Submitted by:

Name: Istiak Alam

ID: 0692230005101005

Batch: CSE-20

Submission Date: April 25, 2026

Submitted to:

A. H. M. Saiful Islam

Chairman, Dept of CSE

Notre Dame University Bangladesh

Table of Contents

1 Implementing Naive Bayes Theorem **1**

2 Naive Bayes Classifier - Titanic **3**
2.1 Calculate value with cross validation 12

3 Feature-selection-stdm **12**
3.1 Univariate Selecetion Method @Feature Selection, Algorithm: SelectKBest Algorithm 12

4 Feature-selection-using-correlation **16**
4.1 Encoding / Handling another column => location 18

5 Feature Selection using Correlation -U **19**
5.1 Encoding / Handling another column => location 20
5.2 Encoding / Handling another column => area 21
5.3 Encoding more than one column at a time 21

1 Implementing Naive Bayes Theorem

[1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report
```

[2]:
```
# Load the dataset
df = pd.read_csv('NaiveBayesDataset.csv')
```

[3]:
```
df
```

[3]:

	Day	Discount	Free_Delivery	Purchase
0	Weekday	Yes	Yes	Yes
1	Weekday	Yes	Yes	Yes
2	Weekday	No	No	No
3	Holiday	Yes	Yes	Yes
4	Weekend	Yes	Yes	Yes
5	Holiday	No	No	No
6	Weekend	Yes	No	Yes
7	Weekday	Yes	Yes	Yes
8	Weekend	Yes	Yes	Yes
9	Holiday	Yes	Yes	Yes
10	Holiday	No	Yes	Yes
11	Holiday	No	No	No
12	Weekend	Yes	Yes	Yes
13	Holiday	Yes	Yes	Yes
14	Holiday	Yes	Yes	Yes
15	Weekday	Yes	Yes	Yes
16	Holiday	No	Yes	Yes
17	Weekday	Yes	No	Yes
18	Weekend	No	No	Yes
19	Weekend	No	Yes	Yes
20	Weekday	Yes	Yes	Yes
21	Weekend	Yes	Yes	No
22	Holiday	No	Yes	Yes
23	Weekday	Yes	Yes	Yes
24	Holiday	No	No	No
25	Weekday	No	Yes	No
26	Weekday	Yes	Yes	Yes
27	Weekday	Yes	Yes	Yes
28	Holiday	Yes	Yes	Yes
29	Weekend	Yes	Yes	Yes

[4]:
```
# Encode categorical variables
label_encoders = {}
for column in ['Day', 'Discount', 'Free_Delivery', 'Purchase']:
    le = LabelEncoder()
    df[column] = le.fit_transform(df[column])
    label_encoders[column] = le
```

[5]:
```python
# Split data into features (X) and target (y)
X = df[['Day', 'Discount', 'Free_Delivery']]
y = df['Purchase']
```

[6]:
```python
# Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)
```

[7]:
```python
# Initialize and train the Naive Bayes model
nb_model = CategoricalNB()
nb_model.fit(X_train, y_train)
```

[7]:
```
CategoricalNB()
```

[8]:
```python
# Make predictions on the test set
y_pred = nb_model.predict(X_test)
```

[9]:
```python
# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
              precision    recall  f1-score   support

           1       1.00      1.00      1.00         6

    accuracy                           1.00         6
   macro avg       1.00      1.00      1.00         6
weighted avg       1.00      1.00      1.00         6
```

[10]:
```python
# Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['No']})
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
prediction = nb_model.predict(example)
print("Purchase Prediction:", label_encoders['Purchase'].␣
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

[11]:
```python
# Check if "Holiday" exists in the Day encoder classes
print("Classes for Day:", label_encoders['Day'].classes_)
```

```
Classes for Day: ['Holiday' 'Weekday' 'Weekend']
```

[12]:
```python
# Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['No']})
# This line creates a DataFrame named example with a single row representing␣
↪an example input.
```

[13]:
```
# Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and␣
↪transforms the categorical
#text values into numeric codes, which the model requires as input.
```

[14]:
```
# Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a␣
↪prediction
# on the transformed example input.
```

[15]:
```
# Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

[16]:
```
# Example prediction
example = pd.DataFrame({'Day': ['Weekend'], 'Discount': ['Yes'],␣
↪'Free_Delivery': ['Yes']})
# This line creates a DataFrame named example with a single row representing␣
↪an example input
```

[17]:
```
# Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and␣
↪transforms the categorical
#text values into numeric codes, which the model requires as input.
```

[18]:
```
# Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a␣
↪prediction
# on the transformed example input.
```

[19]:
```
# Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↪inverse_transform(prediction))
```

```
Purchase Prediction: ['Yes']
```

2 Naive Bayes Classifier - Titanic

[20]:
```
import pandas as pd
```

[21]:
```
df= pd.read_csv('titanic.csv')
df.head()
```

[21]:
```
   PassengerId  Survived  Pclass  \
0          892         0       3
1          893         1       3
2          894         0       2
3          895         0       3
4          896         1       3

                                           Name     Sex   Age  SibSp  Parch ␣
 ↪\
0                              Kelly, Mr. James    male  34.5      0      0
1              Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                     Myles, Mr. Thomas Francis    male  62.0      0      0
3                              Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1

    Ticket     Fare Cabin Embarked
0   330911   7.8292   NaN        Q
1   363272   7.0000   NaN        S
2   240276   9.6875   NaN        Q
3   315154   8.6625   NaN        S
4  3101298  12.2875   NaN        S
```

[22]:
```
df.isnull().sum()
```

[22]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age             86
SibSp            0
Parch            0
Ticket           0
Fare             1
Cabin          327
Embarked         0
dtype: int64
```

[23]:
```
# removing null values of Age// better if we remove later on the updated␣
 ↪dataset
handle = df['Age'].mean()
handle
```

[23]:
```
np.float64(30.272590361445783)
```

[24]:
```
df.Age= df.Age.fillna(handle)
```

[25]:
```
df
```

[25]:
```
     PassengerId  Survived  Pclass  \
0            892         0       3
```

```
1             893         1       3
2             894         0       2
3             895         0       3
4             896         1       3
..            ...       ...     ...
413          1305         0       3
414          1306         1       1
415          1307         0       3
416          1308         0       3
417          1309         0       3

                                             Name     Sex       Age  SibSp  \
0                                Kelly, Mr. James    male  34.50000      0
1                Wilkes, Mrs. James (Ellen Needs)  female  47.00000      1
2                       Myles, Mr. Thomas Francis    male  62.00000      0
3                                Wirz, Mr. Albert    male  27.00000      0
4    Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.00000      1
..                                            ...     ...       ...    ...
413                            Spector, Mr. Woolf    male  30.27259      0
414                  Oliva y Ocana, Dona. Fermina  female  39.00000      0
415                  Saether, Mr. Simon Sivertsen    male  38.50000      0
416                           Ware, Mr. Frederick    male  30.27259      0
417                      Peter, Master. Michael J    male  30.27259      1

     Parch              Ticket      Fare Cabin Embarked
0        0              330911    7.8292   NaN        Q
1        0              363272    7.0000   NaN        S
2        0              240276    9.6875   NaN        Q
3        0              315154    8.6625   NaN        S
4        1             3101298   12.2875   NaN        S
..     ...                 ...       ...   ...      ...
413      0           A.5. 3236    8.0500   NaN        S
414      0            PC 17758  108.9000  C105        C
415      0  SOTON/O.Q. 3101262    7.2500   NaN        S
416      0              359309    8.0500   NaN        S
417      1                2668   22.3583   NaN        C

[418 rows x 12 columns]
```

[26]:
```
# Assumption: Make a naive assumption that features such as␣
↪male,class,age,cabin,fare, etc. are independant of each other
```

[27]:
```
df.
↪drop(['PassengerId','Name','SibSp','Parch','Ticket','Cabin','Embarked'],axis␣
↪='columns',inplace= True)
```

[28]:
```
df.head()
```

[28]:
```
   Survived  Pclass     Sex   Age     Fare
0         0       3    male  34.5   7.8292
1         1       3  female  47.0   7.0000
```

```
2          0        2     male  62.0    9.6875
3          0        3     male  27.0    8.6625
4          1        3   female  22.0   12.2875
```

[29]:
```
# 1 - survived 0 - not survived , At the end we will work on final
# obtained from inputs and targer for train_test_split()
```

[30]:
```
# 1 - survived 0 - not survived
inputs= df.drop('Survived', axis = 'columns')
target = df.Survived
```

[31]:
```
# Encoding Sex
dummies = pd.get_dummies(df['Sex'])
```

[32]:
```
dummies = dummies.astype(int)
# will display o and 1 instead of True and False
```

[33]:
```
dummies
```

[33]:
```
     female  male
0         0     1
1         1     0
2         0     1
3         0     1
4         1     0
..      ...   ...
413       0     1
414       1     0
415       0     1
416       0     1
417       0     1

[418 rows x 2 columns]
```

[34]:
```
#inputs = inputs.drop('Sex','male' axis='columns')
```

[35]:
```
df
```

[35]:
```
     Survived  Pclass     Sex       Age      Fare
0           0       3    male  34.50000    7.8292
1           1       3  female  47.00000    7.0000
2           0       2    male  62.00000    9.6875
3           0       3    male  27.00000    8.6625
4           1       3  female  22.00000   12.2875
..        ...     ...     ...       ...       ...
413         0       3    male  30.27259    8.0500
414         1       1  female  39.00000  108.9000
415         0       3    male  38.50000    7.2500
416         0       3    male  30.27259    8.0500
417         0       3    male  30.27259   22.3583
```

```
[418 rows x 5 columns]
```

[36]:
```
inputs
```

[36]:
```
     Pclass     Sex       Age      Fare
0         3    male  34.50000    7.8292
1         3  female  47.00000    7.0000
2         2    male  62.00000    9.6875
3         3    male  27.00000    8.6625
4         3  female  22.00000   12.2875
..      ...     ...       ...       ...
413       3    male  30.27259    8.0500
414       1  female  39.00000  108.9000
415       3    male  38.50000    7.2500
416       3    male  30.27259    8.0500
417       3    male  30.27259   22.3583

[418 rows x 4 columns]
```

[37]:
```
inputs = inputs.drop(['Sex'] ,axis='columns')
```

[38]:
```
inputs
```

[38]:
```
     Pclass       Age      Fare
0         3  34.50000    7.8292
1         3  47.00000    7.0000
2         2  62.00000    9.6875
3         3  27.00000    8.6625
4         3  22.00000   12.2875
..      ...       ...       ...
413       3  30.27259    8.0500
414       1  39.00000  108.9000
415       3  38.50000    7.2500
416       3  30.27259    8.0500
417       3  30.27259   22.3583

[418 rows x 3 columns]
```

[39]:
```
#concatening columns with sex column
df
```

[39]:
```
     Survived  Pclass     Sex       Age      Fare
0           0       3    male  34.50000    7.8292
1           1       3  female  47.00000    7.0000
2           0       2    male  62.00000    9.6875
3           0       3    male  27.00000    8.6625
4           1       3  female  22.00000   12.2875
..        ...     ...     ...       ...       ...
413         0       3    male  30.27259    8.0500
414         1       1  female  39.00000  108.9000
415         0       3    male  38.50000    7.2500
```

```
416           0       3    male  30.27259    8.0500
417           0       3    male  30.27259   22.3583

[418 rows x 5 columns]
```

[40]:
```
inputs
```

[40]:
```
     Pclass       Age      Fare
0         3  34.50000    7.8292
1         3  47.00000    7.0000
2         2  62.00000    9.6875
3         3  27.00000    8.6625
4         3  22.00000   12.2875
..      …         …        …
413       3  30.27259    8.0500
414       1  39.00000  108.9000
415       3  38.50000    7.2500
416       3  30.27259    8.0500
417       3  30.27259   22.3583

[418 rows x 3 columns]
```

[41]:
```
merged = pd.concat([inputs,dummies],axis = 'columns')
```

[42]:
```
merged
```

[42]:
```
     Pclass       Age      Fare  female  male
0         3  34.50000    7.8292       0     1
1         3  47.00000    7.0000       1     0
2         2  62.00000    9.6875       0     1
3         3  27.00000    8.6625       0     1
4         3  22.00000   12.2875       1     0
..      …         …        …       …     …
413       3  30.27259    8.0500       0     1
414       1  39.00000  108.9000       1     0
415       3  38.50000    7.2500       0     1
416       3  30.27259    8.0500       0     1
417       3  30.27259   22.3583       0     1

[418 rows x 5 columns]
```

[43]:
```
final = merged.drop(['male'],axis ='columns')
```

[44]:
```
final
```

[44]:
```
     Pclass       Age      Fare  female
0         3  34.50000    7.8292       0
1         3  47.00000    7.0000       1
2         2  62.00000    9.6875       0
3         3  27.00000    8.6625       0
4         3  22.00000   12.2875       1
```

```
..      …       …        …      …
413      3  30.27259    8.0500      0
414      1  39.00000  108.9000      1
415      3  38.50000    7.2500      0
416      3  30.27259    8.0500      0
417      3  30.27259   22.3583      0

[418 rows x 4 columns]
```

[45]:
```
final.columns[final.isna().any()]
```

[45]:
```
Index(['Fare'], dtype='str')
```

[46]:
```
null_count_city = df['Fare'].isnull().sum()
print(f"Number of null values in 'Fare' column: {null_count_city}")
```

```
Number of null values in 'Fare' column: 1
```

[47]:
```
final.Fare[:50]
```

[47]:
```
0        7.8292
1        7.0000
2        9.6875
3        8.6625
4       12.2875
5        9.2250
6        7.6292
7       29.0000
8        7.2292
9       24.1500
10       7.8958
11      26.0000
12      82.2667
13      26.0000
14      61.1750
15      27.7208
16      12.3500
17       7.2250
18       7.9250
19       7.2250
20      59.4000
21       3.1708
22      31.6833
23      61.3792
24     262.3750
25      14.5000
26      61.9792
27       7.2250
28      30.5000
29      21.6792
30      26.0000
31      31.5000
```

```
32     20.5750
33     23.4500
34     57.7500
35      7.2292
36      8.0500
37      8.6625
38      9.5000
39     56.4958
40     13.4167
41     26.5500
42      7.8500
43     13.0000
44     52.5542
45      7.9250
46     29.7000
47      7.7500
48     76.2917
49     15.9000
Name: Fare, dtype: float64
```

[48]:
```
final.Fare = final.Fare.fillna(final.Fare.mean())
final.head()
```

[48]:
```
   Pclass   Age     Fare  female
0       3  34.5   7.8292       0
1       3  47.0   7.0000       1
2       2  62.0   9.6875       0
3       3  27.0   8.6625       0
4       3  22.0  12.2875       1
```

[49]:
```
from sklearn.model_selection import train_test_split
```

[50]:
```
xtrain, xtest, ytrain, ytest = train_test_split(final, target, test_size = 0.
↪30, random_state =1)
```

[51]:
```
from sklearn.naive_bayes import GaussianNB
model = GaussianNB ()
```

[52]:
```
model.fit(xtrain, ytrain)
```

[52]:
```
GaussianNB()
```

[53]:
```
model.score(xtest,ytest)
```

[53]:
```
1.0
```

[54]:
```
xtest[:10]
```

[54]:
```
     Pclass       Age     Fare  female
358       3  30.27259   7.7500       0
164       2  41.00000  13.0000       0
```

```
17         3  21.00000    7.2250       0
67         1  47.00000   42.4000       0
4          3  22.00000   12.2875       1
377        2  21.00000   11.5000       0
214        3  38.00000    7.7750       1
290        1  30.27259   39.6000       0
381        3  26.00000    7.8792       0
5          3  14.00000    9.2250       0
```

[55]:
```
ytest[0:10]
```

[55]:
```
358    0
164    0
17     0
67     0
4      1
377    0
214    1
290    0
381    0
5      0
Name: Survived, dtype: int64
```

[56]:
```
model.predict(xtest[0:10])
```

[56]:
```
array([0, 0, 0, 0, 1, 0, 1, 0, 0, 0])
```

[57]:
```
model.predict_proba(xtest[:10])
```

[57]:
```
array([[1., 0.],
       [1., 0.],
       [1., 0.],
       [1., 0.],
       [0., 1.],
       [1., 0.],
       [0., 1.],
       [1., 0.],
       [1., 0.],
       [1., 0.]])
```

[58]:
```
#pclass, Age, Fare, Gender

test = [[1,23.000000,113.2750,0]]
#test = xtest
a= model.predict(test)
if a[0] == 0:
    print("Not Survived")
else:
    print(" Survived")
```

```
Not Survived
```

```
/media/spyder/Secondary-Disk/Github/Student-Performance-Prediction-with-
Explainable-AI/CodeBase/Trial/.spenv/lib/python3.11/site-
packages/sklearn/base.py:493: UserWarning: X does not have valid feature
↪names,
but GaussianNB was fitted with feature names
  warnings.warn(
```

2.1 Calculate value with cross validation

[59]:
```
from sklearn.model_selection import cross_val_score
```

[60]:
```
cross_val_score (GaussianNB(),xtrain,ytrain,cv = 5)
```

[60]:
```
array([1., 1., 1., 1., 1.])
```

3 Feature-selection-stdm

3.1 Univariate Selecetion Method @Feature Selection, Algorithm: SelectKBest Algorithm

[61]:
```
import pandas as pd
import numpy as np
import seaborn as sns
```

[62]:
```
#df = pd.read_csv("bank.csv",sep = ';')
df = pd.read_csv("cardio_train.csv", sep = ';')
```

[63]:
```
df
```

[63]:
```
          id    age  gender  height  weight  ap_hi  ap_lo  cholesterol  gluc
↪ \
0          0  18393       2     168    62.0    110     80            1     1
1          1  20228       1     156    85.0    140     90            3     1
2          2  18857       1     165    64.0    130     70            3     1
3          3  17623       2     169    82.0    150    100            1     1
4          4  17474       1     156    56.0    100     60            1     1
…        …    …       …      …       …      …      …          …     …
69995  99993  19240       2     168    76.0    120     80            1     1
69996  99995  22601       1     158   126.0    140     90            2     2
69997  99996  19066       2     183   105.0    180     90            3     1
69998  99998  22431       1     163    72.0    135     80            1     2
69999  99999  20540       1     170    72.0    120     80            2     1

       smoke  alco  active  cardio
0          0     0       1       0
1          0     0       1       1
2          0     0       0       1
3          0     0       1       1
4          0     0       0       0
…        …     …     …       …
```

```
69995       1      0        1        0
69996       0      0        1        1
69997       0      1        0        1
69998       0      0        0        1
69999       0      0        1        0

[70000 rows x 13 columns]
```

[64]:
```
df.shape
```

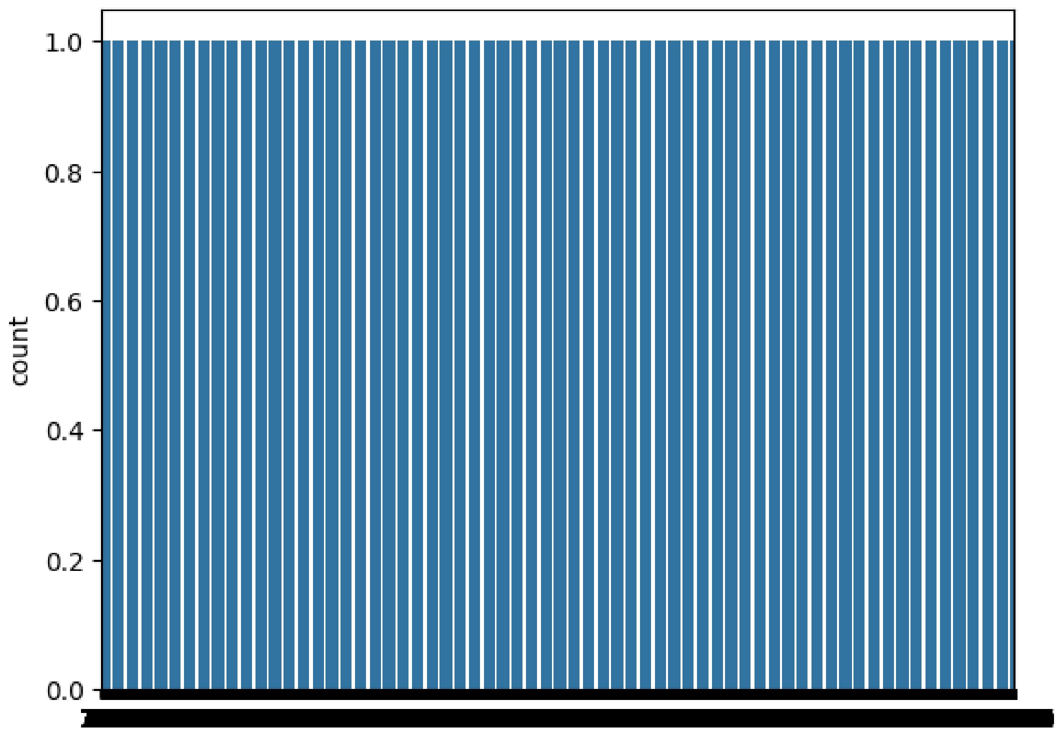
[64]:
```
(70000, 13)
```

[65]:
```
df["cardio"].value_counts()
```

[65]:
```
cardio
0    35021
1    34979
Name: count, dtype: int64
```

[66]:
```
sns.countplot(df["cardio"])
```

[66]:
```
<Axes: ylabel='count'>
```



- x = df.iloc[:, :-1]:
- This line is selecting all rows (:) and all columns except the last one (:-1).
- : means “select all rows.”

- :-1 means "select all columns except the last one." In Python slicing, :-1 excludes the last item, so this will include all columns up to, but not ncluding, the last one.
- Result: x will be a DataFrame containing all the columns of df except the last column

[67]:
```
x = df.iloc[: , :-1]
```

- y = df.iloc[:, 12]
- This line is selecting all rows (:) and the 13th column (remember, indexing in Python starts at 0, so index 12 corresponds to the 13th column).
- : means "select all rows."
- 12 means "select the 13th column" (as Python uses 0-based indexing).
- Result: y will be a Series containing the values of the 13th column of df.

[68]:
```
# y = df.iloc[:, 12]
y = df.iloc [: ,12]
```

[69]:
```
from sklearn.feature_selection import SelectKBest
```

[70]:
```
from sklearn.feature_selection import f_classif
```

[71]:
```
FIT_FEATURES= SelectKBest(score_func = f_classif)
```

[72]:
```
FIT_FEATURES.fit(x,y)
```

[72]:
```
SelectKBest()
```

[73]:
```
pd.DataFrame(FIT_FEATURES.scores_)
```

[73]:

	0
0	1.010461
1	4209.007957
2	4.603641
3	8.197397
4	2388.777887
5	208.339524
6	303.629011
7	3599.361137
8	562.772977
9	16.790541
10	3.761355
11	89.091494

[74]:
```
score_COL = pd.DataFrame(FIT_FEATURES.scores_)
```

[75]:
```
score_COL
```

[75]:

	0
0	1.010461
1	4209.007957
2	4.603641
3	8.197397
4	2388.777887

```
5      208.339524
6      303.629011
7     3599.361137
8      562.772977
9       16.790541
10       3.761355
11      89.091494
```

[76]:
```
# Assuming FIT_FEATURES_scores is an array or a list of scores
score_COL = pd.DataFrame(FIT_FEATURES.scores_, columns=["scorevalue"])
# Now score_COL is a DataFrame with a single column named "scorevalue"
```

[77]:
```
score_COL
```

[77]:
```
     scorevalue
0      1.010461
1   4209.007957
2      4.603641
3      8.197397
4   2388.777887
5    208.339524
6    303.629011
7   3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

[78]:
```
NEW_COL = pd.DataFrame(x.columns)
```

[79]:
```
NEW_COL
```

[79]:
```
              0
0            id
1           age
2        gender
3        height
4        weight
5         ap_hi
6         ap_lo
7   cholesterol
8          gluc
9         smoke
10         alco
11       active
```

[80]:
```
top_features = pd.concat([NEW_COL, score_COL], axis=1)
```

[81]:
```
top_features
```

[81]:

```
               0   scorevalue
0             id     1.010461
1            age  4209.007957
2         gender     4.603641
3         height     8.197397
4         weight  2388.777887
5          ap_hi   208.339524
6          ap_lo   303.629011
7    cholesterol  3599.361137
8           gluc   562.772977
9          smoke    16.790541
10          alco     3.761355
11        active    89.091494
```

[82]:

```
top_features.nlargest (8,"scorevalue")
```

[82]:

```
               0   scorevalue
1            age  4209.007957
7    cholesterol  3599.361137
4         weight  2388.777887
8           gluc   562.772977
6          ap_lo   303.629011
5          ap_hi   208.339524
11        active    89.091494
9          smoke    16.790541
```

4 Feature-selection-using-correlation

[83]:

```
import pandas as pd
# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

[84]:

```
df
```

[84]:

```
  location      area   bedroom   bathroom owner  blodd group  price
0        a     dhaka         3          2     x           o+  10000
1        b     dhaka         4          2     y          ab+  12000
2        c    khulna         2          2     p          ab-  11000
3        d  rajshahi         3          2     q           o+  15000
4        e   cumilla         3          1     w          ab+  12000
5        f     dhaka         2          1     l          ab-  11000
6        g  barishal         2          1     r           o+  10000
7        h    natore         2          3     s           o+  15000
8        i    khulna         3          3     k          ab+  12000
9        j     dhaka         4          2     v          ab-  11000
```

[85]:

```
# Check for non-numeric data in the columns
print(df.dtypes)
```

```
location          str
area              str
```

```
bedroom          int64
bathroom         int64
owner              str
blodd group        str
price            int64
dtype: object
```

[86]:
```
# Identify non-numeric values in specific columns
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

[87]:
```
# Option 1: Convert columns to numeric, forcing errors to NaN
df = df.apply(pd.to_numeric, errors='coerce')
```

[88]:
```
df
```

[88]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	NaN	NaN	3	2	NaN	NaN	10000
1	NaN	NaN	4	2	NaN	NaN	12000
2	NaN	NaN	2	2	NaN	NaN	11000
3	NaN	NaN	3	2	NaN	NaN	15000
4	NaN	NaN	3	1	NaN	NaN	12000
5	NaN	NaN	2	1	NaN	NaN	11000
6	NaN	NaN	2	1	NaN	NaN	10000
7	NaN	NaN	2	3	NaN	NaN	15000
8	NaN	NaN	3	3	NaN	NaN	12000
9	NaN	NaN	4	2	NaN	NaN	11000

[89]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

[90]:
```
print(df.dtypes)
```

```
location       float64
area           float64
bedroom          int64
bathroom         int64
owner          float64
blodd group    float64
price            int64
dtype: object
```

[91]:
```
# Option 1: Convert columns to numeric, forcing errors to NaN
df = df.apply(pd.to_numeric, errors='coerce')
```

[92]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

[93]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
 price          1.000000
bathroom       0.495798
bedroom       -0.015721
location            NaN
area                NaN
owner               NaN
blodd group         NaN
Name: price, dtype: float64
```

4.1 Encoding / Handling another column => location

[94]:
```
from sklearn.preprocessing import OrdinalEncoder
# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

[95]:
```
categorical_column = 'location'
```

[96]:
```
df
```

[96]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[97]:
```
print(df.columns)
```

```
Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
       'price'],
      dtype='str')
```

[98]:
```
df.columns = df.columns.str.strip() # Remove any leading/trailing spaces
df.columns = df.columns.str.lower() # Convert all column names to lowercase
 ↪(optional)
print(df.columns)
```

```
Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
       'price'],
      dtype='str')
```

[99]:
```
# Optional: Use a conditional check
if 'location' in df.columns:
    print(df['location'].head())
else:
    print("The 'location' column is not in the DataFrame.")
```

```
0    a
1    b
2    c
3    d
4    e
Name: location, dtype: str
```

[100]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

[101]:
```
df
```

[101]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	dhaka	3	2	x	o+	10000
1	1.0	dhaka	4	2	y	ab+	12000
2	2.0	khulna	2	2	p	ab-	11000
3	3.0	rajshahi	3	2	q	o+	15000
4	4.0	cumilla	3	1	w	ab+	12000
5	5.0	dhaka	2	1	l	ab-	11000
6	6.0	barishal	2	1	r	o+	10000
7	7.0	natore	2	3	s	o+	15000
8	8.0	khulna	3	3	k	ab+	12000
9	9.0	dhaka	4	2	v	ab-	11000

5 Feature Selection using Correlation -U

[102]:
```
import pandas as pd
# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

[103]:
```
df
```

[103]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[104]:
```
#  Check for non-numeric data in the columns
print(df.dtypes)
```

```
location        str
area            str
bedroom       int64
bathroom      int64
```

```
owner            str
blodd group      str
price          int64
dtype: object
```

[105]:
```
# Identify non-numeric values in specific columns
for column in df.columns:
     if df[column].dtype == 'object':
         print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

5.1 Encoding / Handling another column => location

[106]:
```
from sklearn.preprocessing import OrdinalEncoder
categorical_column = 'location'
df
```

[106]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

[107]:
```
print(df.columns)
```

```
Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
       'price'],
      dtype='str')
```

[108]:
```
df.columns = df.columns.str.strip() # Remove any leading/trailing spaces
df.columns = df.columns.str.lower() # Convert all column names to lowercase␣
 ↪(o
print(df.columns)
```

```
Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
       'price'],
      dtype='str')
```

[109]:
```
# Optional: Use a conditional check
if 'location' in df.columns:
 print(df['location'].head())
else:
 print("The 'location' column is not in the DataFrame.")
```

```
0    a
1    b
2    c
3    d
```

```
4    e
Name: location, dtype: str
```

[110]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

[111]:
```
df
```

[111]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	dhaka	3	2	x	o+	10000
1	1.0	dhaka	4	2	y	ab+	12000
2	2.0	khulna	2	2	p	ab-	11000
3	3.0	rajshahi	3	2	q	o+	15000
4	4.0	cumilla	3	1	w	ab+	12000
5	5.0	dhaka	2	1	l	ab-	11000
6	6.0	barishal	2	1	r	o+	10000
7	7.0	natore	2	3	s	o+	15000
8	8.0	khulna	3	3	k	ab+	12000
9	9.0	dhaka	4	2	v	ab-	11000

5.2 Encoding / Handling another column => area

[112]:
```
categorical_column = 'area'
```

[113]:
```
encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

[114]:
```
df
```

[114]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	x	o+	10000
1	1.0	2.0	4	2	y	ab+	12000
2	2.0	3.0	2	2	p	ab-	11000
3	3.0	5.0	3	2	q	o+	15000
4	4.0	1.0	3	1	w	ab+	12000
5	5.0	2.0	2	1	l	ab-	11000
6	6.0	0.0	2	1	r	o+	10000
7	7.0	4.0	2	3	s	o+	15000
8	8.0	3.0	3	3	k	ab+	12000
9	9.0	2.0	4	2	v	ab-	11000

5.3 Encoding more than one column at a time

[115]:
```
categorical_columns = ['owner', 'blodd group']
```

[116]:
```
for column in categorical_columns:
    if column not in df.columns:
        print(f"Column '{column}' does not exist in the DataFrame")
    else:
        print(f"Encoding column: {column}")
```

```
Encoding column: owner
Encoding column: blodd group
```

[117]:
```
encoder = OrdinalEncoder()
existing_categorical_columns = [col for col in categorical_columns if col in␣
↪df]
df[existing_categorical_columns] = encoder.
↪fit_transform(df[existing_categorical_columns])
```

[118]:
```
df
```

[118]:

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	8.0	2.0	10000
1	1.0	2.0	4	2	9.0	0.0	12000
2	2.0	3.0	2	2	2.0	1.0	11000
3	3.0	5.0	3	2	3.0	2.0	15000
4	4.0	1.0	3	1	7.0	0.0	12000
5	5.0	2.0	2	1	1.0	1.0	11000
6	6.0	0.0	2	1	4.0	2.0	10000
7	7.0	4.0	2	3	5.0	2.0	15000
8	8.0	3.0	3	3	0.0	0.0	12000
9	9.0	2.0	4	2	6.0	1.0	11000

[119]:
```
# Calculate correlation with the target variable 'price'
correlation = df.corr()
```

[120]:
```
# Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
 price          1.000000
area           0.797921
bathroom       0.495798
blodd group    0.148712
location       0.133118
bedroom       -0.015721
owner         -0.092159
Name: price, dtype: float64
```